

SSILP Math

Beginner

LAWS OF LOGARITHMS
EXPONENTIAL & LOG EQUATIONS



Learning Objectives

- Understand the fundamental laws of logarithms and their applications.
- Learn how to use the laws of logarithms to evaluate and simplify expressions.
- Gain proficiency in expanding and combining logarithmic expressions.
- Identify common errors to avoid when working with logarithms.
- Master the change of base formula for logarithms.
- Solve exponential equations by understanding the properties of exponential functions.
- Solve logarithmic equations and understand their properties.
- Tackle word problems involving exponential and logarithmic equations.

Session Outline

1 Laws of Logarithms

2 Exponential and Logarithmic
equations

1 Laws of Logarithms

Laws of logarithms

LAWS OF LOGARITHMS

Let a be a positive number, with $a \neq 1$. Let A , B , and C be any real numbers with $A > 0$ and $B > 0$.

Law

1. $\log_a(AB) = \log_a A + \log_a B$

Description

The logarithm of a product of numbers is the sum of the logarithms of the numbers.

2. $\log_a\left(\frac{A}{B}\right) = \log_a A - \log_a B$

The logarithm of a quotient of numbers is the difference of the logarithms of the numbers.

3. $\log_a(A^C) = C \log_a A$

The logarithm of a power of a number is the exponent times the logarithm of the number.

Using the Laws of Logarithms to Evaluate Expressions

Evaluate each expression.

(a) $\log_4 2 + \log_4 32$

(b) $\log_2 80 - \log_2 5$

(c) $-\frac{1}{3} \log 8$

SOLUTION

$$\begin{aligned} \text{(a)} \quad \log_4 2 + \log_4 32 &= \log_4(2 \cdot 32) \\ &= \log_4 64 = 3 \end{aligned}$$

Law 1

Because $64 = 4^3$

$$\begin{aligned} \text{(b)} \quad \log_2 80 - \log_2 5 &= \log_2\left(\frac{80}{5}\right) \\ &= \log_2 16 = 4 \end{aligned}$$

Law 2

Because $16 = 2^4$

$$\begin{aligned} \text{(c)} \quad -\frac{1}{3} \log 8 &= \log 8^{-1/3} \\ &= \log\left(\frac{1}{2}\right) \end{aligned}$$

Law 3

Property of negative exponents

EXAMPLE

Expanding Logarithmic expressions

Use the Laws of Logarithms to expand each expression.

(a) $\log_2(6x)$ (b) $\log_5(x^3y^6)$ (c) $\ln\left(\frac{ab}{\sqrt[3]{c}}\right)$

SOLUTION

(a) $\log_2(6x) = \log_2 6 + \log_2 x$ Law 1

(b) $\log_5(x^3y^6) = \log_5 x^3 + \log_5 y^6$ Law 1
 $= 3 \log_5 x + 6 \log_5 y$ Law 3

(c) $\ln\left(\frac{ab}{\sqrt[3]{c}}\right) = \ln(ab) - \ln \sqrt[3]{c}$ Law 2
 $= \ln a + \ln b - \ln c^{1/3}$ Law 1
 $= \ln a + \ln b - \frac{1}{3} \ln c$ Law 3

Combining Logarithmic expressions

Use the Laws of Logarithms to combine each expression into a single logarithm.

(a) $3 \log x + \frac{1}{2} \log(x + 1)$

(b) $3 \ln s + \frac{1}{2} \ln t - 4 \ln(t^2 + 1)$

SOLUTION

$$\begin{aligned} \text{(a)} \quad 3 \log x + \frac{1}{2} \log(x + 1) &= \log x^3 + \log(x + 1)^{1/2} && \text{Law 3} \\ &= \log(x^3(x + 1)^{1/2}) && \text{Law 1} \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad 3 \ln s + \frac{1}{2} \ln t - 4 \ln(t^2 + 1) &= \ln s^3 + \ln t^{1/2} - \ln(t^2 + 1)^4 && \text{Law 3} \\ &= \ln(s^3 t^{1/2}) - \ln(t^2 + 1)^4 && \text{Law 1} \\ &= \ln\left(\frac{s^3 \sqrt{t}}{(t^2 + 1)^4}\right) && \text{Law 2} \end{aligned}$$

EXAMPLE

Logarithms Common errors to avoid

Warning Although the Laws of Logarithms tell us how to compute the logarithm of a product or a quotient, *there is no corresponding rule for the logarithm of a sum or a difference*. For instance,

$$\log_a(x + y) \not\equiv \log_a x + \log_a y$$

In fact, we know that the right side is equal to $\log_a(xy)$. Also, don't improperly simplify quotients or powers of logarithms. For instance,

$$\frac{\log 6}{\log 2} \not\equiv \log\left(\frac{6}{2}\right) \quad \text{and} \quad (\log_2 x)^3 \not\equiv 3 \log_2 x$$

Change of base formula

Suppose we are given $\log_a x$ and want to find $\log_b x$. Let

$$y = \log_b x$$

We write this in exponential form and take the logarithm, with base a , of each side.

$$b^y = x \quad \text{Exponential form}$$

$$\log_a(b^y) = \log_a x \quad \text{Take } \log_a \text{ of each side}$$

$$y \log_a b = \log_a x \quad \text{Law 3}$$

$$y = \frac{\log_a x}{\log_a b} \quad \text{Divide by } \log_a b$$

CHANGE OF BASE FORMULA

$$\log_b x = \frac{\log_a x}{\log_a b}$$

In particular, if we put $x = a$, then $\log_a a = 1$, and this formula becomes

$$\log_b a = \frac{1}{\log_a b}$$

Use Laws of Logarithms to evaluate

(a) $\log_3 135 - \log_3 45$

(b) $-\frac{1}{3} \log_3 27$

(c) $\log_3 100 - \log_3 18 - \log_3 50$

ACTIVITY 1.1

1.1 Use Laws of Logarithms to evaluate

Expanding Logarithmic expressions

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(a) $\log_3(x\sqrt{y})$

(b) $\log \sqrt[3]{x^2 + 4}$

(c) $\ln \frac{3x^2}{(x + 1)^{10}}$

ACTIVITY 1.2

1.2 Expanding Logarithmic expressions

(a) $\log_3(x\sqrt{y})$

(b) $\log \sqrt[3]{x^2 + 4}$

SOLUTIONS

1.2 Expanding Logarithmic expressions

(c) $\ln \frac{3x^2}{(x+1)^{10}}$

SOLUTION

Combining Logarithmic expressions

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(a) $\frac{1}{2} \log_2 5 - 2 \log_2 7$

(b) $\log_5(x^2 - 1) - \log_5(x - 1)$

ACTIVITY 1.3

1.3 Combining Logarithmic expressions

(a) $\frac{1}{2} \log_2 5 - 2 \log_2 7$

(b) $\log_5(x^2 - 1) - \log_5(x - 1)$

SOLUTIONS

2 Exponential and Logarithmic equations

Exponential equations

An **exponential equation** is one in which the **variable occurs in the exponent**.

Some exponential equations can be solved by using the fact that exponential functions are **one-to-one**.

$$a^x = a^y \Rightarrow x = y$$

EXAMPLE

Solve the exponential equation.

(a) $5^x = 125$ (b) $5^{2x} = 5^{x+1}$

SOLUTION

(a) We first express 125 as a power of 5 and then use the fact that the exponential function $f(x) = 5^x$ is one-to-one.

$$\begin{array}{ll} 5^x = 125 & \text{Given equation} \\ 5^x = 5^3 & \text{Because } 125 = 5^3 \\ x = 3 & \text{One-to-one property} \end{array}$$

The solution is $x = 3$.

(b) We first use the fact that the function $f(x) = 5^x$ is one-to-one.

$$\begin{array}{ll} 5^{2x} = 5^{x+1} & \text{Given equation} \\ 2x = x + 1 & \text{One-to-one property} \\ x = 1 & \text{Solve for } x \end{array}$$

The solution is $x = 1$.

We solved the equation here by **comparing the exponents** on both sides.

Not suitable for more **complex problems!**

EXAMPLE

Guideline for solving exponential equations

GUIDELINES FOR SOLVING EXPONENTIAL EQUATIONS

1. Isolate the exponential expression on one side of the equation.
2. Take the logarithm of each side, then use the Laws of Logarithms to “bring down the exponent.”
3. Solve for the variable.

Solving an exponential equation

Consider the exponential equation $3^{x+2} = 7$.

(a) Find the exact solution of the equation expressed in terms of logarithms.

We take the common logarithm of each side and use Law 3.

$$3^{x+2} = 7 \quad \text{Given equation}$$

$$\log(3^{x+2}) = \log 7 \quad \text{Take log of each side}$$

$$(x + 2)\log 3 = \log 7 \quad \text{Law 3 (bring down exponent)}$$

$$x + 2 = \frac{\log 7}{\log 3} \quad \text{Divide by } \log 3$$

$$x = \frac{\log 7}{\log 3} - 2 \quad \text{Subtract 2}$$

The exact solution is $x = \frac{\log 7}{\log 3} - 2$.

Solving an exponential equation

Solve the equation $8e^{2x} = 20$.

SOLUTION We first divide by 8 to isolate the exponential term on one side of the equation.

$$8e^{2x} = 20 \quad \text{Given equation}$$

$$e^{2x} = \frac{20}{8} \quad \text{Divide by 8}$$

$$\ln e^{2x} = \ln 2.5 \quad \text{Take } \ln \text{ of each side}$$

$$2x = \ln 2.5 \quad \text{Property of } \ln$$

$$x = \frac{\ln 2.5}{2} \quad \text{Divide by 2 (exact solution)}$$

$$\approx 0.458 \quad \text{Calculator (approximate solution)}$$

EXAMPLE

Exponential equation of quadratic type

Solve the equation $e^{2x} - e^x - 6 = 0$.

SOLUTION To isolate the exponential term, we factor.

$$e^{2x} - e^x - 6 = 0 \quad \text{Given equation}$$

$$(e^x)^2 - e^x - 6 = 0 \quad \text{Law of Exponents}$$

$$(e^x - 3)(e^x + 2) = 0 \quad \text{Factor (a quadratic in } e^x)$$

$$e^x - 3 = 0 \quad \text{or} \quad e^x + 2 = 0 \quad \text{Zero-Product Property}$$

$$e^x = 3 \qquad e^x = -2$$

The equation $e^x = 3$ leads to $x = \ln 3$. But the equation $e^x = -2$ has no solution because $e^x > 0$ for all x . Thus $x = \ln 3 \approx 1.0986$ is the only solution. You should check that this answer satisfies the original equation.

An Equation Involving Exponential Functions

Solve the equation $3xe^x + x^2e^x = 0$.

SOLUTION First we factor the left side of the equation.

$$3xe^x + x^2e^x = 0 \quad \text{Given equation}$$

$$x(3 + x)e^x = 0 \quad \text{Factor out common factors}$$

$$x(3 + x) = 0 \quad \text{Divide by } e^x \text{ (because } e^x \neq 0)$$

$$x = 0 \quad \text{or} \quad 3 + x = 0 \quad \text{Zero-Product Property}$$

Thus the solutions are $x = 0$ and $x = -3$.

(a) $10^{2x-3} = \frac{1}{10}$

(b) $10^{2x^2-3} = 10^{9-x^2}$

2.1 Exponential equations

(a) $10^{2x-3} = \frac{1}{10}$

(b) $10^{2x^2-3} = 10^{9-x^2}$

SOLUTIONS

Solve in terms of logarithms

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(a) $2e^{12x} = 17$

(b) $1 + e^{4x+1} = 20$

(c) $e^{2x} - e^x - 6 = 0$

ACTIVITY 2.2

2.2 a,b: Solve in terms of logarithms

(a) $2e^{12x} = 17$

(b) $1 + e^{4x+1} = 20$

SOLUTIONS

2.2 c: Solve in terms of logarithms

(c) $e^{2x} - e^x - 6 = 0$

SOLUTION

Logarithmic equations

A *logarithmic equation* is one in which a logarithm of the variable occurs.

GUIDELINES FOR SOLVING LOGARITHMIC EQUATIONS

1. Isolate the logarithmic term on one side of the equation; you might first need to combine the logarithmic terms.
2. Write the equation in exponential form (or raise the base to each side of the equation).
3. Solve for the variable.

Solving logarithmic equations

Solve each equation for x .

(a) $\ln x = 8$

(b) $\log_2(25 - x) = 3$

SOLUTION

(a) $\ln x = 8$ Given equation

$x = e^8$ Exponential form

Therefore $x = e^8 \approx 2981$.

We can also solve this problem another way.

$\ln x = 8$ Given equation

$e^{\ln x} = e^8$ Raise e to each side

$x = e^8$ Property of $\ln$

(b) The first step is to rewrite the equation in exponential form.

$\log_2(25 - x) = 3$ Given equation

$25 - x = 2^3$ Exponential form (or raise 2 to each side)

$25 - x = 8$

$x = 25 - 8 = 17$

EXAMPLE

Solving logarithmic equations

Solve the equation $4 + 3 \log(2x) = 16$.

SOLUTION We first isolate the logarithmic term. This allows us to write the equation in exponential form.

$$4 + 3 \log(2x) = 16 \quad \text{Given equation}$$

$$3 \log(2x) = 12 \quad \text{Subtract 4}$$

$$\log(2x) = 4 \quad \text{Divide by 3}$$

$$2x = 10^4 \quad \text{Exponential form (or raise 10 to each side)}$$

$$x = 5000 \quad \text{Divide by 2}$$

EXAMPLE

Solving logarithmic equations

CHECK YOUR ANSWERS

$x = -4$:

$$\begin{aligned}\log(-4 + 2) + \log(-4 - 1) \\ &= \log(-2) + \log(-5) \\ &\text{undefined} \quad \times\end{aligned}$$

$x = 3$:

$$\begin{aligned}\log(3 + 2) + \log(3 - 1) \\ &= \log 5 + \log 2 = \log(5 \cdot 2) \\ &= \log 10 = 1 \quad \checkmark\end{aligned}$$

Solve the equation $\log(x + 2) + \log(x - 1) = 1$ algebraically and graphically.

SOLUTION 1: Algebraic

We first combine the logarithmic terms, using the Laws of Logarithms.

$$\begin{aligned}\log[(x + 2)(x - 1)] &= 1 && \text{Law 1} \\ (x + 2)(x - 1) &= 10 && \text{Exponential form (or raise 10 to each side)} \\ x^2 + x - 2 &= 10 && \text{Expand left side} \\ x^2 + x - 12 &= 0 && \text{Subtract 10} \\ (x + 4)(x - 3) &= 0 && \text{Factor} \\ x = -4 &\quad \text{or} \quad x = 3\end{aligned}$$

We check these potential solutions in the original equation and find that $x = -4$ is not a solution (because logarithms of negative numbers are undefined), but $x = 3$ is a solution. (See *Check Your Answers*.)

EXAMPLE

Solve in terms of logarithms

(a) $\log_5 x + \log_5(x + 1) = \log_5 20$

(b) $\ln\left(x - \frac{1}{2}\right) + \ln 2 = 2 \ln x$

(c) $\log_2(x^2 - x - 2) = 2$

ACTIVITY 2.3

2.3: Solve in terms of logarithms

(a)

(b)

SOLUTION

2.3: Solve in terms of logarithms

(c)

SOLUTION

- (I) **Fish Population** A small lake is stocked with a certain species of fish. The fish population is modeled by the function

$$P = \frac{10}{1 + 4e^{-0.8t}}$$

where P is the number of fish in thousands and t is measured in years since the lake was stocked.

- (a) Find the fish population after 3 years.
- (b) After how many years will the fish population reach 5000 fish?

- (II) **Atmospheric Pressure** Atmospheric pressure P (in kilopascals, kPa) at altitude h (in kilometers, km) is governed by the formula

$$\ln\left(\frac{P}{P_0}\right) = -\frac{h}{k}$$

where $k = 7$ and $P_0 = 100$ kPa are constants.

- (a) Solve the equation for P .
- (b) Use part (a) to find the pressure P at an altitude of 4 km.

2.4: Solve in terms of logarithms

(I)

(II)

SOLUTION